Automated Backup and Recovery System

Project: Automated Backup and Recovery System

Services: EC2, EBS, Lambda, CloudWatch Events

Description: Automate EBS snapshot creation and deletion using AWS Lambda triggered by CloudWatch Events. Implement snapshot retention policies to ensure old backups are automatically removed.

Skills: Automation, backup strategies, Lambda scripting, IAM troubleshooting

Definition: The Automated Backup and Recovery System is designed to provide a reliable and fully automated backup solution for AWS EC2 instance volumes (EBS). The system uses AWS Lambda to create snapshots of specific EBS volumes and automatically deletes older snapshots based on a defined retention policy. CloudWatch Events (EventBridge) is used to schedule the Lambda function at regular intervals, ensuring consistent and automated backups without manual intervention.

Scope of the Project: The scope of this project is to design and implement a serverless, automated backup system for AWS EC2 EBS volumes. The system ensures data protection, cost optimization, and operational efficiency through automation.

This project covers:

- **Automated EBS Snapshot Creation:** Automatically creates backups of a specified EBS volume.
- **Snapshot Retention Management:**Deletes snapshots older than a defined retention period to optimize storage usage and cost.
- **Scheduled Automation:**Uses CloudWatch Events to trigger backup operations at regular intervals.

Methodology:

1. Lambda + CloudWatch Events (Implemented Method)

- Uses EC2, EBS, Lambda, and CloudWatch Events as per project requirement
- Fully serverless and automated solution
- Lambda handles snapshot creation and deletion logic
- EventBridge triggers Lambda on schedule
- Cost-effective and suitable for free-tier usage
- Provides flexibility through customizable retention policies

2. EC2 + CRON + AWS CLI (Alternative Method)

- Uses EC2 instance with cron jobs and AWS CLI scripts
- Requires continuous EC2 uptime
- Higher operational cost and maintenance
- Suitable for legacy or highly customized environments

AWS Services Used in the Project:

1. Amazon EC2 (Elastic Compute Cloud)

- **Definition:** Provides scalable virtual servers in the cloud
- **Purpose:** Hosts applications and attaches EBS volumes
- **Role:** Source of data; its attached EBS volume is backed up

2. Amazon EBS (Elastic Block Store)

- **Definition:** Persistent block storage service for EC2
- **Purpose:** Stores application data
- **Role:** Snapshots are created from EBS volumes

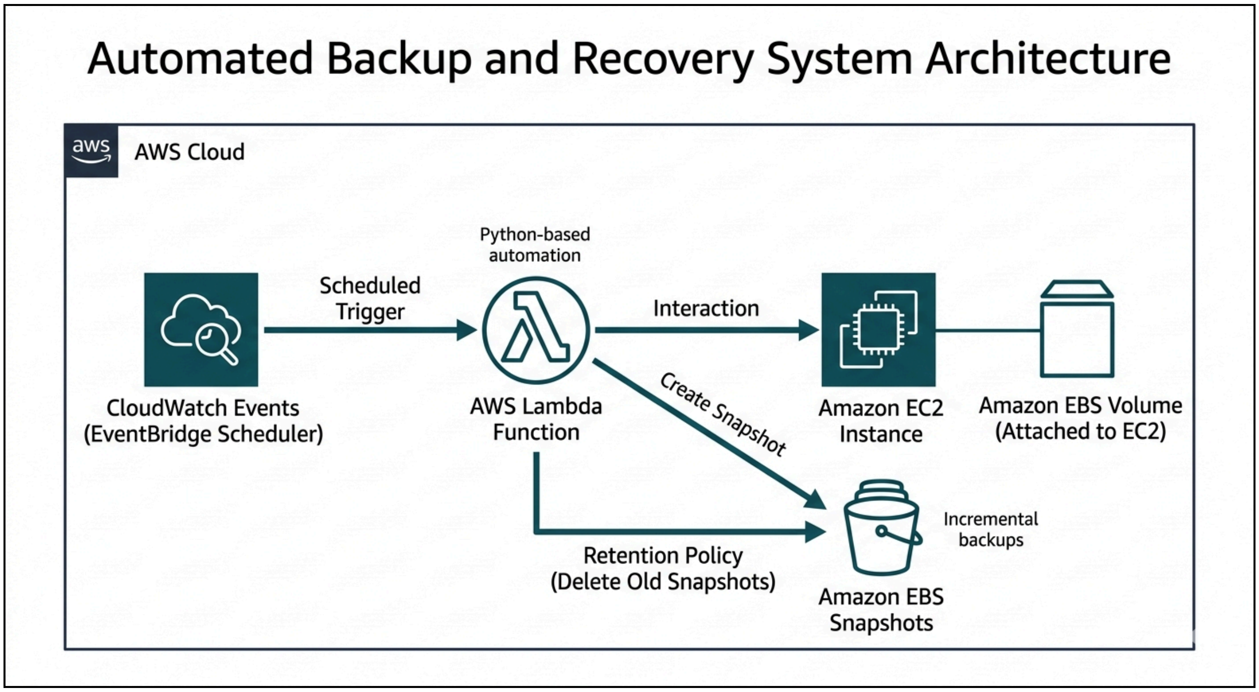
3. AWS Lambda

- **Definition:** Serverless compute service to run code without managing servers
- **Purpose:** Automates snapshot creation and deletion
- **Role:** Creates snapshots of EBS volume, Tags snapshots, Deletes old snapshots based on retention policy

4. Amazon CloudWatch Events (EventBridge)

- **Definition:** Event-driven service for scheduling and automation
- **Purpose:** Triggers Lambda function periodically
- **Role:** Ensures continuous automated backup execution

Architecture Diagram:



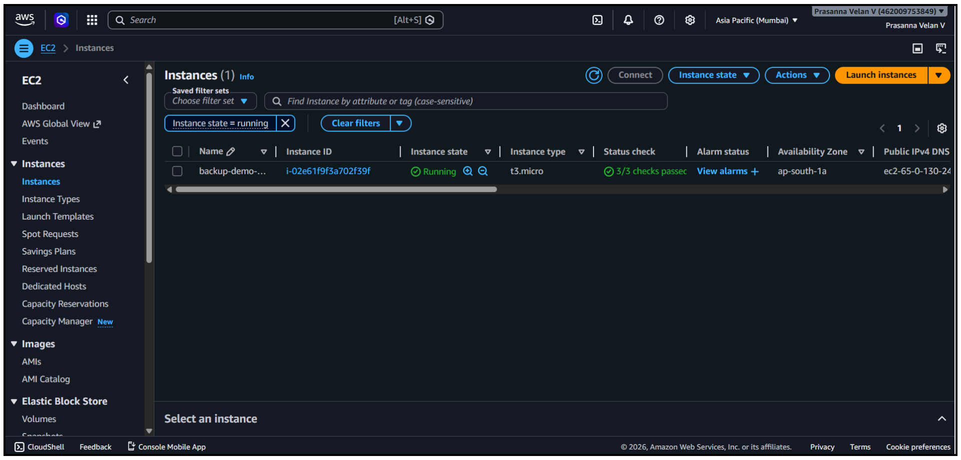
The architecture flow should be visually clear and follow this sequence:

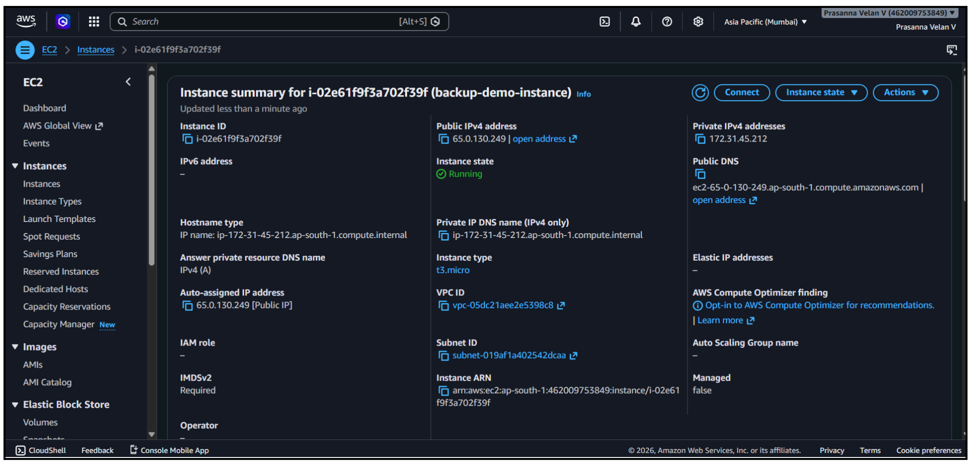
1. CloudWatch Events (EventBridge) triggers the Lambda function on a schedule.
2. The Lambda function interacts with the EBS volume attached to the EC2 instance.
3. Lambda creates EBS snapshots (backup process).
4. Lambda also deletes old snapshots based on a retention policy.

Implementation Steps to Do the Project:

Step 1: Launch EC2 Instance

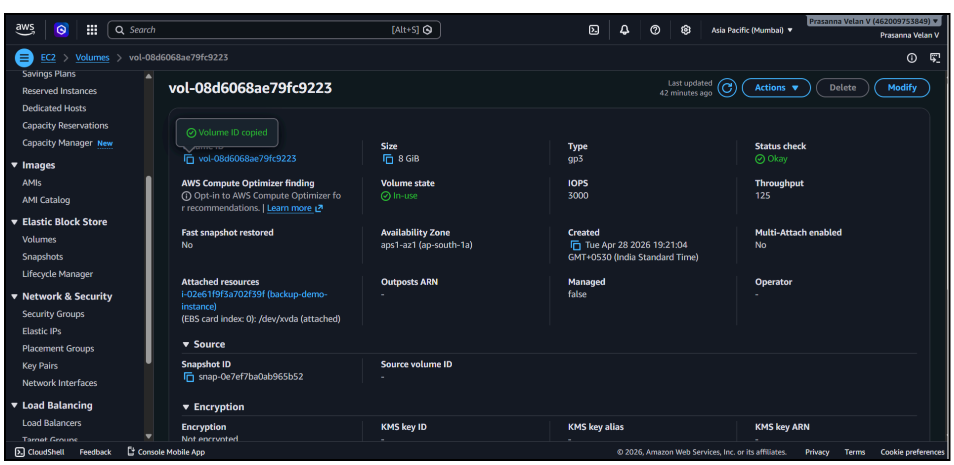
- Go to AWS Console → EC2 → Launch Instance
- Select Amazon Linux (Free Tier)
- Instance type: t2.micro
- Configure storage: 8 GB (default)
- Launch instance





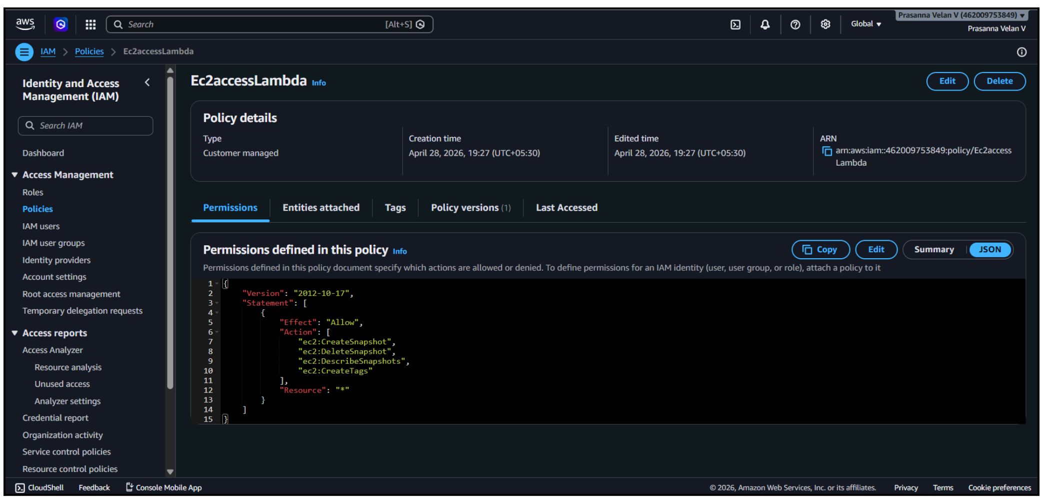
Step 2: Identify EBS Volume

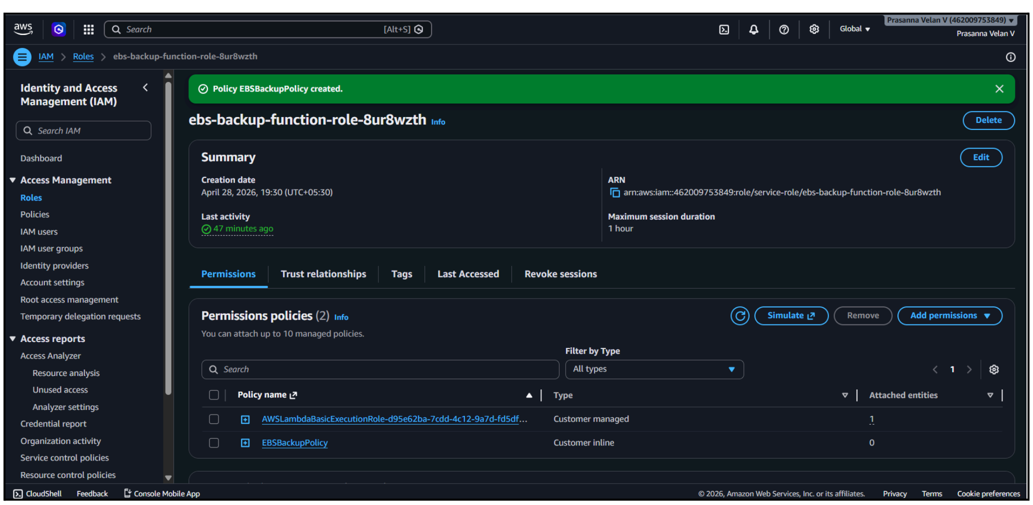
- Go to EC2 → Volumes
- Copy the Volume ID (used in Lambda code)



Step 3: Create IAM Role for Lambda

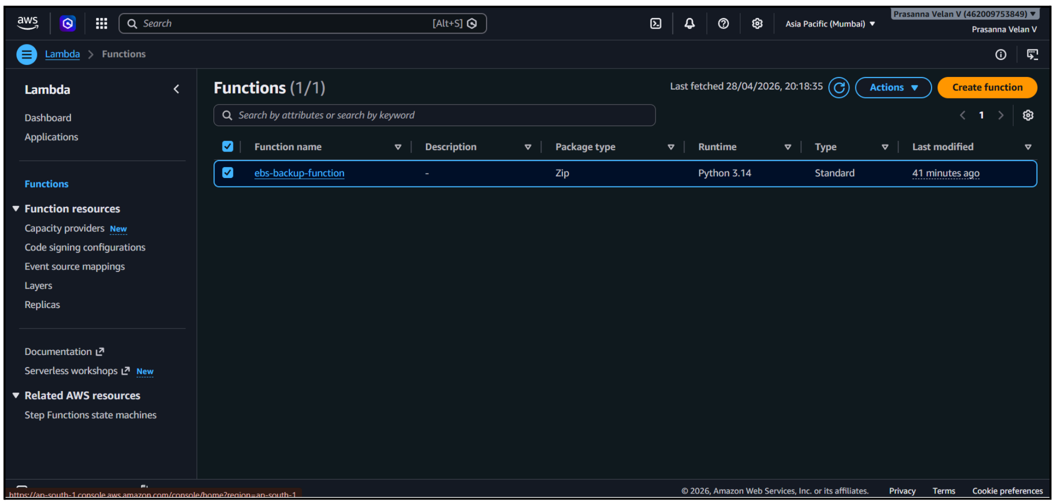
- Go to IAM → Roles → Create Role
- Select AWS Service → Lambda
- Create custom policy with permissions:
    - ec2:CreateSnapshot
    - ec2:DeleteSnapshot
    - ec2:DescribeSnapshots
    - ec2:CreateTags
    - ec2:DescribeVolumes
- Attach policy
- Role name: ebs-backup-function-role





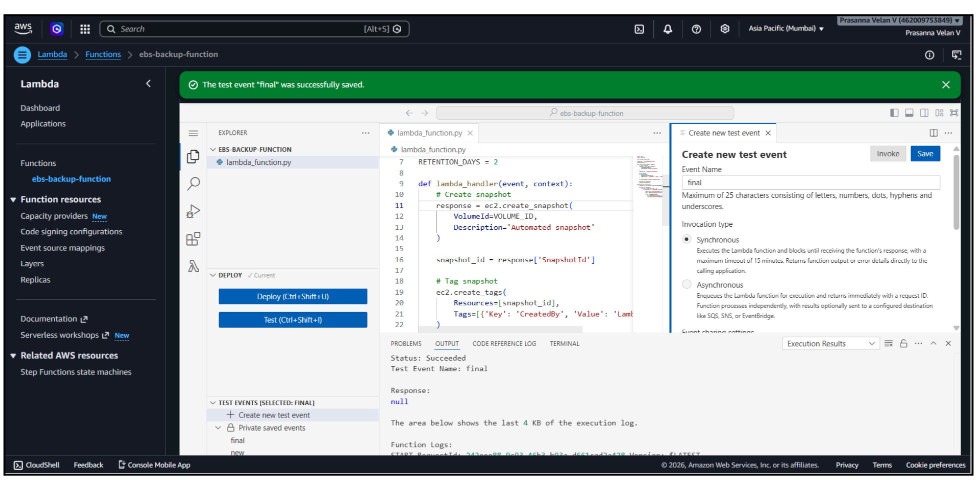
Step 4: Create Lambda Function

- Go to Lambda → Create Function
- Name: ebs-backup-function
- Runtime: Python 3.14
- Attach IAM role



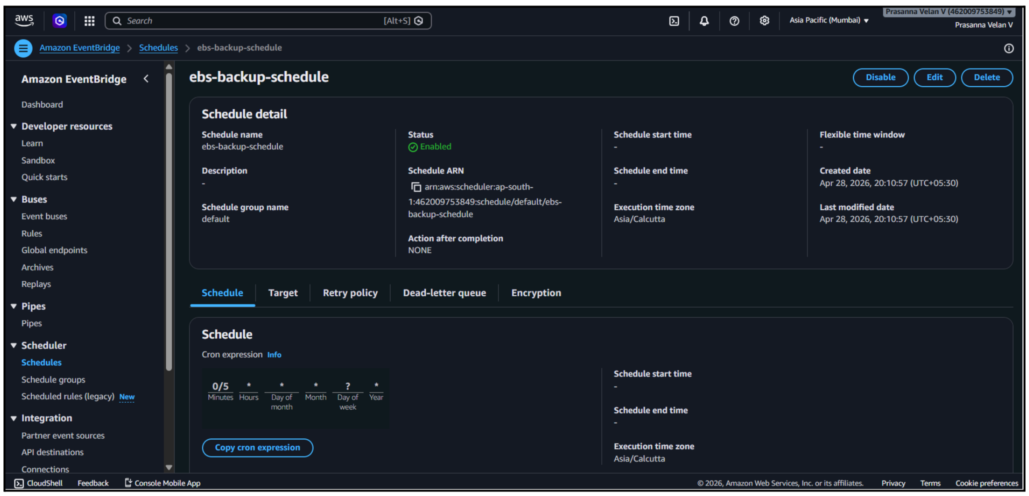
Step 5: Add Lambda Code and Test

- Paste backup script
- Replace Volume ID
- Click Test
- Verify:
  - Go to EC2 → Snapshots
  - Confirm snapshot is created



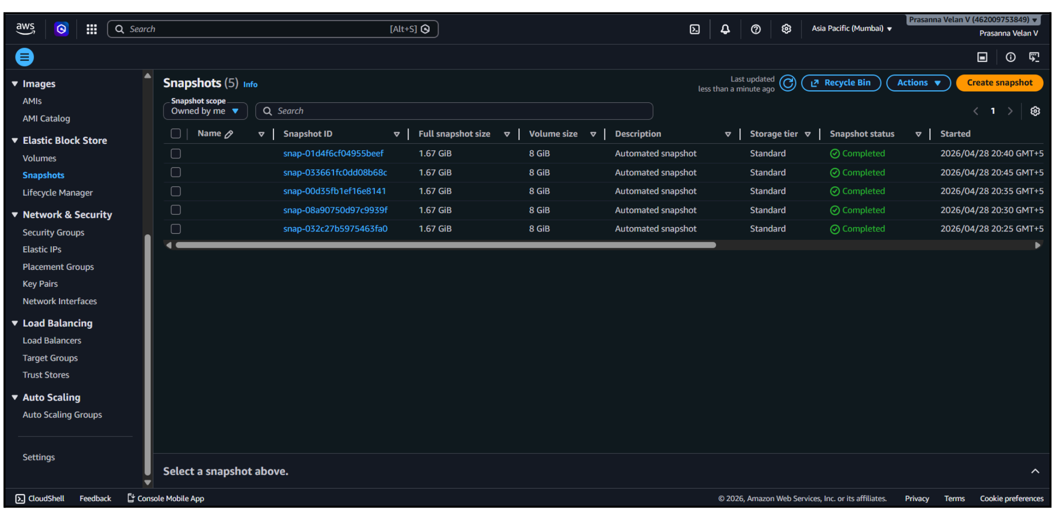
Step 6: Schedule Lambda Using CloudWatch Events

- Go to EventBridge → Create Rule
- Choose Schedule
- Use expression: rate(5 minutes)
- Select Lambda function as target
- Enable rule



Step 7: Verify Snapshots & Retention

- Go to EC2 → Snapshots
- Snapshots are created automatically
- Older snapshots are deleted after retention period



Troubleshooting:

Issue 1: UnauthorizedOperation Error

Lambda failed due to insufficient permissions

Solution:
Attach correct IAM policy with EC2 snapshot permissions

Issue 2: Schedule Not Triggering

No snapshots created automatically

Solution:

- Ensure EventBridge rule is enabled
- Verify Lambda is set as target
- Use rate() instead of incorrect cron

No Snapshots After Schedule

Cron misunderstood (runs at specific time only)

Solution:
Use rate - 5 minutes for testing

Conclusion:

The Automated Backup and Recovery System provides a reliable and efficient solution for protecting EC2 data using EBS snapshots. By leveraging Lambda and CloudWatch Events, the system automates both snapshot creation and retention management, reducing manual effort and operational complexity.

This solution is serverless, scalable, and cost-effective, making it suitable for real-world production environments as well as free-tier projects. It demonstrates strong understanding of AWS automation, IAM permissions, and event-driven architecture.